THE SIX-TERM SEQUENCE FOR INF AND T ON A 2-FIBRE PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3726-3749 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 3751-3783, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 30 of queue_artin.json (pb=1350, stmt 3726-3749, proof 3751-3783).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned. Named omission: the exactness chase is delegated to the formal deformation theory engine lemma - a standard diagram chase; the boundary construction is in-span.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0040: sources/stacks-artin/pinned-original-source.tex, source lines 3783–3783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0041: sources/stacks-artin/pinned-original-source.tex, source lines 3783–3783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06L5: sources/stacks-artin/pinned-original-source.tex, source lines 3783–3783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06I7: sources/stacks-artin/pinned-original-source.tex, source lines 3783–3783. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CXQ: sources/stacks-artin/pinned-original-source.tex, source lines 2922–2928. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07Y7: sources/stacks-artin/pinned-original-source.tex, source lines 3481–3500. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $p : \mathcal{X} \to \mathcal{Y}$ and $q : \mathcal{Z} \to \mathcal{Y}$ be 1-morphisms of categories fibred in groupoids over $(Sch/S)_{fppf}$. Assume $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ satisfy (RS*). Let A be an S-algebra and let w be an object of $\mathcal{W} = \mathcal{X} \times_{\mathcal{Y}} \mathcal{Z}$ over A. Denote x, y, z the objects of $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ you get from w. For any A-module M there is a 6-term exact sequence*

$$\begin{array}{ccccccc} 0 & \longrightarrow & Inf_w(M) & \longrightarrow & Inf_x(M) \oplus Inf_z(M) & \longrightarrow & Inf_y(M) \\ & & & & & \swarrow & \\ & & T_w(M) & \longrightarrow & T_x(M) \oplus T_z(M) & \longrightarrow & T_y(M) \end{array}$$

of A-modules.

Proof. By Lemma 0CXQ we see that $\mathcal{W}$ satisfies (RS*) and hence $T_w(M)$ and $\mathrm{Inf}_w(M)$ are defined. The horizontal arrows are defined using the functoriality of Lemma 07Y7.

Definition of the "boundary" map $\delta : \mathrm{Inf}_y(M) \to T_w(M)$. Choose isomorphisms $p(x) \to y$ and $y \to q(z)$ such that $w = (x, z, p(x) \to y \to q(z))$ in the description of the 2-fibre product of Categories, Lemma 0041 and more precisely Categories, Lemma 0040. Let x', y', z', w' denote the trivial deformation of x, y, z, w over $A[M]$. By pullback we get isomorphisms $y' \to p(x')$ and $q(z') \to y'$. An element $\alpha \in \mathrm{Inf}_y(M)$ is the same thing as an automorphism $\alpha : y' \to y'$ over $A[M]$ which restricts to the identity on y over A. Thus setting

$$\delta(\alpha) = (x', z', p(x') \to y' \xrightarrow{\alpha} y' \to q(z'))$$

we obtain an object of $T_w(M)$. This is a map of A-modules by Formal Deformation Theory, Lemma 06I7.

The rest of the proof is exactly the same as the proof of Formal Deformation Theory, Lemma 06L5. □